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C

Let the amount of added heat be Q .

So,

$$Q = mc_w \Delta T_w \dots\dots\dots (1) \text{ \{heat added to water\}}$$

$$Q = mc_i \Delta T_i \dots\dots\dots (2) \text{ \{heat added to iron\}}$$

$$\text{Eqn. (1) = (2), } \frac{c_w}{c_i} = \frac{\Delta T_i}{\Delta T_w}$$

Since $c_w > c_i$, $\Delta T_i > \Delta T_w \rightarrow$ the final temperature of iron is going to be higher than the water's final temperature.

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